

Status	Finished
Started	Thursday, 9 October 2025, 12:00 PM
Completed	Thursday, 9 October 2025, 12:19 PM
Duration	19 mins 21 secs
Marks	2.00/3.00
Grade	2.00 out of 3.00 (66.68%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1
Partially correct
Mark 0.80 out of 1.00

Consider the following EBNF grammar.

$X \rightarrow o \mid xZ$
 $Z \rightarrow kR \mid vL$
 $L \rightarrow kD \mid oL$
 $D \rightarrow nL \mid oR \mid xR$
 $R \rightarrow X \mid n \mid vX$

Translate it into an NFA, given as 5-tuple.

Use the nonterminal names as the corresponding state names and F for the final state.

Your answers must be syntactically correct for marking.

Sample 5-tuple for an NFA with three states A,B,C with A as start state, C as final state, and three transitions: from A with a to B, from B with b to C, and from C with ε to A:

States = {"A","B","C"}
Alphabet = {"a","b"}
Transitions = { [{"A","a","B"}], [{"B","b","C"}], [{"C","","A"}] }
Start state ="A"
Final States = {"C"}

States= [{"X","Z","L","D","R"}]

Alphabet= [{"o","x","k","v","n"}]

Transitions= [{"X","o","F"}, {"X","x","Z"}, {"Z","k","R"}, {"Z","v","L"}, {"L","k","D"}, {"L","o","L"}, {"D","n","L"}, {"D","o","R"}, {"D","x","R"}, {"R","","X"}, {"R","n","F"}, {"R","v","X"}]

Start state= "X"

Final States= {"F"}

Your last answer was interpreted as follows:

{X, Z, L, D, R}

Your last answer was interpreted as follows:

{o, x, k, v, n}

Your last answer was interpreted as follows:

{[X, o, F], [X, x, Z], [Z, k, R], [Z, v, L], [L, k, D], [L, o, L], [D, n, L], [D, o, R], [D, x, R], [R, , X], [R, n, F], [R, v, X]}

Your last answer was interpreted as follows:

X

Your last answer was interpreted as follows:

{F}

Question **2**
Partially correct
Mark 0.20 out of 1.00

Which temporal logic statement matches the English sentence "It is always the case that a darkness leads to a continuous light."?

F(darkness $\Rightarrow$ G light)

Question **3**
Correct
Mark 1.00 out of 1.00

- Given the set $[1, 2, 3, 4, 5]$, and relations $\{[4, 1], [5, 5], [2, 2], [4, 3], [3, 3], [5, 1]\}$.
1. Fill in the relations matrix below such that all given relations are marked by a "1" in the correct position.
 2. Then extend the set of relations by adding mandatory "1"s so that the final set of relations is a partial order.
 3. All other entries in the matrix must be marked "0".

-	1	2	3	4	5
1	1	0	0	0	0
2	0	1	0	0	0
3	0	0	1	0	0
4	1	0	1	1	0
5	1	0	0	0	1

Your last answer was interpreted as follows:

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
1	0	1	1	0
1	0	0	0	1

Question **4**
Correct
Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.
Tests missing this signing code will be ignored and will not count towards the final score.

Key: 352

Signing code:

Your last answer was interpreted as follows:

48826

➔ [Test 1 \(topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars\)](#)

Jump to...